

Amylase and pH: Continuous-Sampling Investigation · TA brief

TA Brief

Say once: "I haven't seen what you've been doing... I start from what you show me".

P17 · SLT planning and evaluation

Lesson code: SCI-LAUNCH-A1-W08-L2. SoW reference: LAUNCH Science, Autumn 1 return block; L2.

Exact SoW outcome: I can follow the centre-approved Pearson method to investigate how pH affects amylase activity, recording safe continuous-sampling evidence accurately.

Source placement: enzyme activity enrichment consolidates Topic 1; the workbook has no separate Autumn 1 Week 8 row. This is the supplied return-period mapping, not a newly invented workbook objective.

This lesson: I can use continuous-sampling evidence to investigate how pH affects amylase activity.

Evidence (scheme of work): Edexcel 1BI0 Paper 1 (1F) topic tests; AQA UAS science units. F: topic quizzes, practical write-ups. S: Topics 1 to 3 assessment (Dec). Evidence (this lesson): Order the preparation, timing and sampling steps, and justify one evidence decision.; Identify the first observed iodine endpoint and record the time with its sampling limit.; Use matched conditions and raw observations to make a supported comparison.

Evidence location: the pupil's explanation under the book strip, with route, actual support, review edit // and exit response. Keep raw observations distinct from illustrative data.

Label evidence on the school's evidence app with this code: SCI-LAUNCH-A1-W08-L2. Record the observed science and prompt level; clicks and access aids do not earn a marking code.

Stage	Minutes	Task	Staff move
Arrival task	3	Choose a route. Think first, then check the answers together.	At arrival, say the spoken line on the marking card once. Use a truthful last lesson evidence inheritance. Arrival is the only calibration; read the four responses with the starter by minute eight. Choose routes at each task, with no record of switching. IF NEW — 0.5 min read the clues; 1.5 min respond; 1 min sample and reveal. IF REVISION — 0.5 min orient; 1.5 min respond; 1 min sample and reveal.

Stage	Minutes	Task	Staff move
Starter · ready before mixing	4	Setup A: the reaction is mixed, then the group looks for a timer and labels its iodine wells. Setup B: timer, labels and agreed roles are ready before mixing.	Read the arrival to make one reachable adjustment: hold the colour model longer, use the supported We-do prompt, change the current route or give the Aspire evidence card. This changes talk within the existing time. IF NEW — 1 min read the two setups; 1 min choose; 2 min explain the lost evidence. IF REVISION — 1 min predict; 2 min justify; 1 min check the method boundary.
I do · follow the sampled drops	4	The reaction mixture stays separate from the iodine spotting wells. Sample at the agreed fixed interval.	Model only separated samples going into prepared iodine wells; iodine is not poured into the reaction. Teaching timeline is illustrative, not a practical method or evidence of completion. IF NEW — 2.5 min step through the wells; 1.5 min explain the first observed endpoint. IF REVISION — 0.5 min orient; 2.5 min pupils predict each well meaning; 1 min check the timing window.

Stage	Minutes	Task	Staff move
We do · method-decision checkpoint	4	Drag a card, or select it then choose its destination.	Named involvement opportunity. Use card-then-destination by keyboard, touch or pointer; no pointer drag is required. A spill or unidentified solution goes to the adult, not a pupil-designed repair. IF NEW — 1 min read situation and destinations; 2 min place and explain; 1 min respond to a pupil's reasoning. IF REVISION — 0.5 min orient; 2.5 min classify with reasons; 1 min compare the evidence.
I do · record what was observed	4	The first pH 7 endpoint is recorded at 90 s; its threshold crossing lies after 60 and by 90 s. The repeat is 120 s. Keep both raw values. No endpoint by 300 s does not mean no reaction or a zero rate.	Use “after 60 s and by 90 s”. Do not call 75 s the measured endpoint or claim the true time was known to ± 15 s. Sampling uncertainty is not all uncertainty. IF NEW — 2 min model the record; 1 min distinguish observation from inference; 1 min pupil retells. IF REVISION — 1 min inspect; 2 min explain the interval and a control; 1 min compare a missing sample.
We do · repair an evidence claim	5	Observed: blue-black at 60 s; orange-brown at 90 s. A pupil records “endpoint = 75 s exactly”.	A pupil can dictate or point to the repair. Keep gaps and ambiguous colours visible; a smooth expected graph is never a reason to change a value. IF NEW — 1 min read the claim; 3 min repair or direct; 1 min check. IF REVISION — 0.5 min read; 3.5 min repair with a reason; 1 min check.

Stage	Minutes	Task	Staff move
You do · keep a defensible record	10	Use the supplied evidence. Write, draw, point, explain or direct an adult.	Within this twelve-minute block an authorised group may run or record only its assigned trial if the centre method fits; equipment is prepared before class. Use the complete supplied-data route whenever approval, access or time does not fit. Do not run a rushed full pH series or claim paper analysis completes the core practical. IF NEW — 2 min orient the supplied evidence or authorised role; 8 min complete the task; 2 min use one precise feedback point. IF REVISION — 1 min orient; 9 min complete the task; 2 min use one precise feedback point.
Review and improve	3	Keep one correct science claim. Improve one word, unit, piece of evidence or reason. Mark a real change //; read it back if asked.	Give one specific VF and one NS+; wait for a real // edit and hear the pupil explain the change.
Exit ticket	3	Use your book OR one ticket. Show your explanation with the supplied evidence. Say it, write it, draw it, sign it, choose it or show it; an adult can scribe it.	Use one book response OR one ticket. Read once to choose the next teaching move for last lesson's rate calculation and evidence claims. No extra receipt, signature, tally or later evidence-chasing. IF NEW — 1 min read evidence; 2 min respond; 1 min pupil finishes the same response. IF REVISION — 0.5 min orient; 2.5 min respond; 1 min pupil finishes the same response.

Planned adaptations

Need	Planned adaptation	EEF five-a-day strategy
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Cognition and learning	Arrival: reveal one clue at a time; model a reasoning step, then fade the cue for a fresh example.	scaffolding
Communication and interaction	We do: allow thinking time, paired rehearsal, pointing, sign or AAC before giving a reason.	flexible grouping
SEMH	Independent: agree a quiet pause and one return step; correct privately; accept a pass on a personal example.	cognitive and metacognitive strategies
Sensory and physical	Models: offer reduced motion, paused diagrams, enlarged prepared pieces, seated work or adult-directed placement.	technology
EAL (language access, not SEN)	Arrival: pre-teach topic words beside the diagram; permit any-language planning and a sentence stem for the same science.	explicit instruction

Staff: confirm the planned adaptations actually used and record any changes below. EEF strategies are linked to actual tasks above.

If behind

Must teach core: I can use continuous-sampling evidence to investigate how pH affects amylase activity. Shorten or merge: take one shared explanation; preserve independent evidence, review and exit. Next lesson: revisit any insecure criterion before extending the topic. Dependency: Order the preparation, timing and sampling steps, and justify one evidence decision.

Staff evaluation

Date taught	Record actual date.
Delivered as planned	Yes / Partly / No; name change.
Criteria met	Name the criterion and evidence.
Adaptations made and why	Name actual support and effect.
Change from the SoW and coverage impact	State change, reason and next teaching implication.
Evidence banked	Record evidence location and lesson code on the school's evidence app.

P18 · Individual target opportunities

Cognition and learning: Arrival: reveal one clue at a time; model a reasoning step, then fade the cue for a fresh example.

Communication and interaction: We do: allow thinking time, paired rehearsal, pointing, sign or AAC before giving a reason.

SEMH: Independent: agree a quiet pause and one return step; correct privately; accept a pass on a personal example.

Sensory and physical: Models: offer reduced motion, paused diagrams, enlarged prepared pieces, seated work or adult-directed placement.

EAL (language access, not SEN): Arrival: pre-teach topic words beside the diagram; permit any-language planning and a sentence stem for the same science.

Capture E against the pupil's own IEDP target on the school's evidence app; the IEDP carries their EHCP-linked and Boxall targets, so one record serves every plan.

Follow the pupil's IBSP where it differs from these routines.

Adult deployment and responsive feedback

At Independent, one adult supports access to Supported while the other holds the group; swap across the week. Wait, gesture, cue, model one step, then try a fresh step independently. Give VF, then one precise NS+; wait for a real pupil response R and // edit. Use ? for read-back. WS and I describe actual support; E requires target-matched evidence.

During We do, the lesson teacher is the named audience. Offer an optional request for reading help, fewer pieces, clearer labels, adult-directed handling or thinking time. Try the change and check its effect. Passing and private requests are valid.

First lesson back · once-only read

Before the lesson, scan the books for five minutes. Notice previous feedback, pupil responses and where the learning stopped.

Ask the cover teacher and TAs what was actually taught, who was present and what changed for individual pupils.

Check the roll against the books. New arrivals have no prior evidence; start from what every pupil shows now.

Existing NS+ is history. Do not chase, back-mark or retrospectively close it. Begin fresh feedback from this lesson.

After the check, one read, once, shapes Autumn 2. Read each success criterion with its access support, choose the next route, then teach; do not reopen earlier feedback.

Preparation and practical safety

Use the supplied models, cards, data and booklet. This lesson plans only. During the later practical, handle reagents only with the centre-approved method, prepared equipment, local risk assessment and trained supervision. The supplied-observation route needs no reagents and teaches the same interpretation; it does not certify physical practical completion. Do not infer concentrations, heating arrangements or disposal from lesson cards.

Destination link (Launch): the amylase and pH core practical is examined in GCSE Biology; enzyme assays like it are routine in brewing, baking and clinical laboratories, and a Level 3 science or laboratory-technician route builds directly on it.

Choose a route at each task using what the pupil shows. Switching adds no task, record or label.

Risk-note controls cross-checked against publicly accessible CLEAPSS SSS002 Enzymes and SSS056 Iodine on 26 September 2026. Teacher/technician must authorise the local method, concentrations, quantities, PPE and disposal using current member guidance. The full member-only PP048 method and Hazcards were not accessible for this check.

Use minimum enzyme and iodine concentrations and quantities. Prevent enzyme powder aerosol; do not spray solutions. Follow eye-protection and solvent-specific iodine controls.

Illustrative or supplied data is reasoning practice only and is never logged as practical completion. Log only an observed, authorised physical contribution; the teacher decides whether it meets the relevant course requirement.

Teacher answers and topic guidance

Objective mapping: The shared pupil objective retains the pH, amylase and continuous-sampling science. Physical performance against the original objective is assessed only when a pupil actually follows an assigned part of the centre-approved method with authorisation; the complete supplied-data route supports the same interpretation goals but cannot establish practical completion.

Real-world link

Teacher answers

Shared activities

Starter · ready before mixing: Prepare labelled iodine wells and the recording times before starting the authorised reaction. The reaction does not pause while equipment is prepared. Sampled-drop model: 30 s and 60 s: blue-black, starch detected. 90 s: orange-brown, starch not detected. Record first observed

endpoint 90 s; it was reached after 60 s and by 90 s. Method-decision checkpoint: Continue as authorised: ready checked tile; clean fixed sample. Stop/check adult: iodine spill; unlabelled buffer; temperature outside the authorised range. Preserve evidence and seek an authorised repeat/check: missed interval; borderline colour. Never invent a missing observation. Record and repair: Replace “75 s exactly” with “first observed endpoint 90 s; reached after 60 s and by 90 s”. A missed 60 s sample widens the known interval from after 30 s to by 90 s. Keep the missed result visible.

Supported arrival

1. Amylase is the enzyme and speeds up starch breakdown.
2. Yes, starch is detected in the sampled drop.
3. Endpoint time is measured; buffer pH is changed.
4. Prepare the iodine wells before mixing and starting timing.

Supported independent task

1. 90 s; starch is not detected in that sampled drop.
2. Prepare and check → start together → sample at fixed times → keep the raw observation. Recording occurs as sampling proceeds; the last card names the evidence habit throughout.
3. Temperature stays matched. pH is the variable changed; matching temperature helps isolate the pH comparison.
4. First observed endpoint = 90 s. The exact moment between 60 s and 90 s is unknown.
5. pH 7 is faster in this series: 90 s compared with 240 s at pH 4, under the stated matched conditions.

Supported exit

1. 90 s; starch is not detected in that sample.
2. Record 90 s. We did not observe 75 s; the endpoint happened after 60 s and by 90 s.

Independent arrival

1. Amylase is the enzyme; starch is the substrate.
2. 90 s. The exact change between the two samples was not observed.
3. Independent: buffer pH. Dependent: endpoint time. A valid control is temperature, solution volume/concentration or sampling interval.
4. Preparation allows samples and observations to begin at the agreed times, reducing missed early observations.

Independent independent task

1. First observed endpoint = 90 s; it was reached after 60 s and by 90 s. Blue-black at 60 s shows starch still detected then.
2. Prepare/check → start → sample → record honestly as sampling proceeds. Preparation avoids losing observations while the reaction continues.
3. IV: buffer pH. DV: endpoint time. Keep temperature or starting amounts/volumes/concentrations matched, so their changes do not offer another explanation.
4. Record 90 s as the first observed endpoint, not an invented 75 s. Record the missed sample explicitly and retain actual later observations.
5. For example pH 7: 90 s, versus pH 4: 240 s, so pH 7 is faster for these conditions. One trial at each of five pH values cannot identify an exact universal optimum.

Independent exit

1. Record 90 s; the endpoint was reached after 60 s and by 90 s.
2. For example, keep temperature matched so a temperature change cannot explain a difference attributed to pH.

Aspire arrival

1. The colour indicates whether starch remains; its disappearance under controlled conditions supports breakdown, rather than directly measuring molecular activity.
2. No. Record 90 s as the first observed endpoint; the endpoint was reached after 60 s and by 90 s.
3. Matched temperature limits an alternative explanation across pH trials; the no-active-amylase comparison checks for change without enzyme activity.
4. No. If 30 s was positive and 90 s negative, the observed window is wider: after 30 s and by 90 s. Preserve the missed sample.

Aspire independent task

1. Record first observed endpoint = 90 s. The endpoint was reached after 60 s and by 90 s, a 30 s window; its exact moment is unknown.
2. Preparation prevents a delay at the start; fixed intervals give comparable sampling opportunities and a stated timing limit. Neither removes all colour-judgement uncertainty.
3. Matched temperature prevents a competing temperature effect across the pH trials. The no-active-amylase comparison checks whether starch disappearance requires active enzyme under the same conditions.
4. The endpoint was reached after 30 s and by 90 s, a 60 s window. Record 60 s as missed and the first observed endpoint as 90 s; the timing precision is reduced.
5. The pH 7 endpoint is earlier (90 s versus 240 s) under matched conditions. This first series supports faster breakdown at pH 7 than pH 4 here. Raw observations must remain visible; uncertainty calls for checking or an authorised repeat, not a preferred value.

Aspire exit

1. Still record first observed endpoint = 90 s; the window is after 30 s and by 90 s, widening from 30 s to 60 s.
2. A shorter fixed interval, if authorised and feasible, narrows the timing window. It does not remove subjective judgement of borderline iodine colour.

OpenStax: enzymes

Next teaching after last lesson: Autumn 2, after half term.

LAUNCH SCIENCE · last lesson · Week 8 · Autumn 2

Staff copy — not for pupil books

Lead with VF in the practical and exit tasks. Written feedback is for extended writing only, strategic and not every piece (Curriculum Policy 2026/27 §14.3).

Core practical evidence. The Edexcel core practical is evidenced by the pupil doing the practical. The illustrative route evidences the interpretation skills only, and is never recorded as core practical completion.

Code

Meaning

Verbal feedback given here

Worked with support

NS + ...

Next step one specific thing

Evidence captured on the school's digital evidence platform

Responded loop closed (Audience has happened)

Self-edit point

Read this back to me

Curriculum Policy 2026/27 §14.3 summary: VF WS I NS E R (six codes; NS “next step”, E “evidence on digital software app”, R “responded to feedback”). The Feedback Policy, the code source, adds // and ? and expands NS to one specific thing and R to loop closed.

Voice + Audience = R: the pupil responds AND the adult genuinely engages with that response; add R only when both have happened. For a margin edit, the policy says: “in the same colour, on the same page”.

Supported: Use the colour key to repair one endpoint statement: first observed time, colour and starch meaning.

Independent: Replace your exact midpoint with the first observed endpoint and its after/by time window.

Aspire: Repair one uncertainty claim using the actual last positive sample, first negative sample and an improvement that targets that gap.

Secure: In last lesson connect the same endpoints to relative rate and a bounded graph claim.

Mixed: Work one endpoint statement beside its colour key before calculating a reciprocal.

Misconception: Contrast the last starch-positive sample with the first negative sample; an unobserved midpoint is not a measurement.

Access barrier: Use the large labelled wells and allow a pupil-directed spoken or pointed explanation.

Method-or-data problem: Re-establish actual sampled times, preserve gaps and use the supplied intact dataset for analysis.

We do · method-decision checkpoint. Invite a pupil to choose a situation's destination and explain the evidence; respond to that reasoning. VF records verbal feedback actually given, not participation alone.

In our enzyme-planning lesson, your evidence showed ____, so today we're starting with ____.

LAUNCH: "Adult engages with the edit against the mark scheme / model answer before R." The pupil owns the response; no adult counter-initial or extra receipt. Codes stay in books. No second next step or retrospective closure.

Answer: Amylase is the enzyme and speeds up starch breakdown.

Answer: Yes, starch is detected in the sampled drop.

Answer: Endpoint time is measured; buffer pH is changed.

Answer: Prepare the iodine wells before mixing and starting timing.

Answer: Amylase is the enzyme; starch is the substrate.

Answer: 90 s. The exact change between the two samples was not observed.

Answer: Independent: buffer pH. Dependent: endpoint time. A valid control is temperature, solution volume/concentration or sampling interval.

Answer: Preparation allows samples and observations to begin at the agreed times, reducing missed early observations.

Answer: The colour indicates whether starch remains; its disappearance under controlled conditions supports breakdown, rather than directly measuring molecular activity.

Answer: No. Record 90 s as the first observed endpoint; the endpoint was reached after 60 s and by 90 s.

Answer: Matched temperature limits an alternative explanation across pH trials; the no-active-amylase comparison checks for change without enzyme activity.

Answer: No. If 30 s was positive and 90 s negative, the observed window is wider: after 30 s and by 90 s. Preserve the missed sample.

Science staff guidance

Beside the You do, mark I (independent), S (supported) or C (choice): the level of independence, not the effort.

Give time to start → point to the relevant place → rephrase → offer a small prompt or word bank → reduce to a limited choice. Never straight to telling the answer.

No cold calling; nobody is asked to read difficult text aloud.

Tell the teacher the barrier: reading, vocabulary, processing, regulation, writing or a science misconception. Avoid prompting through every question or dictating the final answer.

E Evidence on the school's evidence app

Lean capture on the school's evidence app (E): 30-second clip, photo, or one-line context note.

Point, sign, repeat after the adult, demonstrate, a clip on the school's evidence app, choice board.

If behind

A pupil who missed last lesson: give the Arrival reminders and today's must-know words, not the missed lesson's work.

Photograph the practical at the planned point, upload it to the school's evidence app with this code, and stick a copy in the book.

Planned point: after the observation is recorded. Lesson code: SCI-LAUNCH-A1-W08-L2. Only the apparatus for the current stage is on the table. Use the teacher demonstration plus the same observation and conclusion task if a pupil cannot safely take part.

Arrival and Starter

Arrival: answer these four short questions in the book.

1. Identify the enzyme and substrate in amylase acting on starch.
2. At 60 s iodine is blue-black; at 90 s it stays orange-brown. Which sampled time first meets the endpoint?
3. State the independent and dependent variables, then name one condition to keep constant.
4. Why prepare the iodine wells and time headings before mixing?

Starter: Which setup protects the first observations? One short answer in the book.

You do routes

Independent is preferred: work in the book without answer-giving support. Supported uses a small prompt beside the same task. Choice uses limited options, matching or selecting and then a short scientific sentence, label or reason in the book. Aspire is the challenge for every route.

Mark scheme · one point per mark**Independent**

1. State the first observed endpoint for pH 7 and the interval in which it was reached, using the supplied observations. [2 marks]
 1. Record 90 s (1 mark)
 2. The crossing was after 60 and by 90 s (1 mark)
2. Order the four method cards. Explain why preparation must happen before mixing. [2 marks]
 1. Prepare/check, start together, sample at fixed times, record raw observations (1 mark)
 2. Reaction starts on mixing, so preparation must be completed first (1 mark)
3. State the independent variable, dependent variable and one matched control, explaining why the control matters. [4 marks]
 1. Change pH (1 mark)
 2. Measure endpoint time (1 mark)
 3. Keep temperature or solution amounts matched (1 mark)
 4. The named control could affect rate independently of pH (1 mark)
4. Repair the proposed exact endpoint of 75 s. Explain what should be recorded if a sample is missed. [2 marks]
 1. 75 s was not observed, so record 90 s (1 mark)
 2. Keep a missed-sample gap and use actual observed bounds (1 mark)
5. Compare two pH values using their times. State one limit of the conclusion from this first series. [3 marks]
 1. For example pH 7 takes 90 s and pH 4 takes 240 s (1 mark)
 2. pH 7 is faster in these matched first runs (1 mark)
 3. Single trials, tested range or subjective endpoint limits the claim (1 mark)

Aspire

1. Distinguish the recorded endpoint from the true endpoint window for pH 7. State the window's width. [3 marks]
 1. Record 90 s (1 mark)
 2. Threshold crossing is after 60 and by 90 s (1 mark)
 3. The window is 30 s wide (1 mark)
2. Explain how preparing the tile first and using a fixed sampling interval protect different parts of the evidence. [2 marks]
 1. Preparing the tile avoids losing early observations (1 mark)
 2. A fixed sampling interval gives consistent timing gaps (1 mark)
3. Explain the different roles of matched temperature and a comparison containing no active amylase. [2 marks]
 1. Matching temperature limits its effect as an alternative cause (1 mark)
 2. The no-active-amylase comparison tests change without active enzyme (1 mark)
4. Evaluate this claim: The 60 s sample is missing in a second record; 30 s is blue-black and 90 s orange-brown. State what is now known and repair the claim "the same precision". [3 marks]
 1. The observed crossing is after 30 and by 90 s (1 mark)
 2. The window is 60 s wide (1 mark)
 3. It is wider than the original 30 s window (1 mark)
5. Compare pH 4 and pH 7 using the supplied times, limiting your claim to these trials. Explain why neither a smooth expected trend nor a hard-to-read colour justifies altering a raw observation. [3 marks]
 1. pH 7 has a shorter endpoint time than pH 4: 90 versus 240 s (1 mark)
 2. The conclusion is limited to these matched trials (1 mark)
 3. Changing observations to fit an expected trend would make the record untruthful (1 mark)

Accept scientifically equivalent wording. Award each distinct point once. The mark tariff is for the question, not a participation score. Keep the answer scheme staff-only during independent work.